# CSGE602055 Operating Systems
# CSF2600505 Sistem Operasi
# Week 06: Concurrency: Processes & Threads

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https://os.vlsm.org/Slides/os06.pdf
Always check for the latest revision!

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# OS221[3]): Operating Systems Schedule 2022 - 1

| Week | Topic[1]) | OSC10[2]) |
|---|---|---|
| Week 00 | Overview 1, Virtualization & Scripting | Ch. 1, 2, 18. |
| Week 01 | Overview 2, Virtualization & Scripting | Ch. 1, 2, 18. |
| Week 02 | Security, Protection, Privacy, & C-language. | Ch. 16, 17. |
| Week 03 | File System & FUSE | Ch. 13, 14, 15. |
| Week 04 | Addressing, Shared Lib, & Pointer | Ch. 9. |
| Week 05 | Virtual Memory | Ch. 10. |
| Week 06 | Concurrency: Processes & Threads | Ch. 3, 4. |
| Week 07 | Synchronization & Deadlock | Ch. 6, 7, 8. |
| Week 08 | Scheduling + W06/W07 | Ch. 5. |
| Week 09 | Storage, Firmware, Bootloader, & Systemd | Ch. 11. |
| Week 10 | I/O & Programming | Ch. 12. |

[1]) For schedule, see https://os.vlsm.org/#idx02

[2]) Silberschatz et. al.: **Operating System Concepts**, $10^{th}$ Edition, 2018.

[3]) This information will be on **EVERY** page two (2) of this course material.

# STARTING POINT — https://os.vlsm.org/

- **Text Book** — Any recent/decent OS book. Eg. (**OSC10**) Silberschatz et. al.: **Operating System Concepts**, $10^{th}$ Edition, 2018. See also https://www.os-book.com/OS10/.
- **Resources**
  - **SCELE OS221** — https://scele.cs.ui.ac.id/course/view.php?id=3313. The enrollment key is **XXX**.
  - **Download Slides and Demos from GitHub.com** https://github.com/os2xx/os/
    os00.pdf (W00), os01.pdf (W01), os02.pdf (W02), os03.pdf (W03), os04.pdf (W04), os05.pdf (W05), os06.pdf (W06), os07.pdf (W07), os08.pdf (W08), os09.pdf (W09), os10.pdf (W10).
  - **Problems**
    195.pdf (W00), 196.pdf (W01), 197.pdf (W02), 198.pdf (W03), 199.pdf (W04), 200.pdf (W05), 201.pdf (W06), 202.pdf (W07), 203.pdf (W08), 204.pdf (W09), 205.pdf (W10).
  - **LFS** — http://www.linuxfromscratch.org/lfs/view/stable/
  - **OSP4DISS** — https://osp4diss.vlsm.org/
  - **DOIT** — https://doit.vlsm.org/001.html

# Agenda


1. Start
2. OS221 Schedule
3. Agenda
4. Week 06
5. Week 01: Review 2

# Agenda (2)

# Week 06 Concurrency: Topics[1]

- States and state diagrams
- Structures (ready list, process control blocks, and so forth)
- Dispatching and context switching
- The role of interrupts
- Managing atomic access to OS objects
- Implementing synchronization primitives
- Multiprocessor issues (spin-locks, reentrancy)

[1] Source: ACM IEEE CS Curricula 2013

# Week 06 Concurrency: Learning Outcomes (1)[1]

- Describe the need for concurrency within the framework of an operating system. [Familiarity]
- Demonstrate the potential run-time problems arising from the concurrent operation of many separate tasks. [Usage]
- Summarize the range of mechanisms that can be employed at the operating system level to realize concurrent systems and describe the benefits of each. [Familiarity]
- Explain the different states that a task may pass through and the data structures needed to support the management of many tasks. [Familiarity]

[1] Source: ACM IEEE CS Curricula 2013

# Week 06 Concurrency: Learning Outcomes (2)[1]

- Summarize techniques for achieving synchronization in an operating system (e.g., describe how to implement a semaphore using OS primitives). [Familiarity]
- Describe reasons for using interrupts, dispatching, and context switching to support concurrency in an operating system. [Familiarity]
- Create state and transition diagrams for simple problem domains. [Usage]

[1] Source: ACM IEEE CS Curricula 2013

# The Operating System

- TBA